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Cybercrime Education in U.S. Criminology and Criminal Justice Doctoral Programs: Curricula, Faculty Capacity, and Interdisciplinary Challenges

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ABSTRACT

Cybercrime has become an increasingly important area within criminology and criminal justice, yet little is known about how doctoral programs prepare scholars to research and teach in this domain. This study examines cybercrime education in U.S. criminology and criminal justice PhD programs by compiling and analyzing a database of cybercrime-related courses and faculty expertise across 43 institutions. We assess curricular design and faculty capacity supporting cybercrime-focused training. Results show that about half of the programs offer at least one cybercrime-related course, but only six maintain dedicated cybercrime laboratories. Cybercrime instruction is typically elective rather than core, and opportunities for technical training are limited. Faculty specializing in cybercrime are increasing but remain largely early career, with nearly half trained in liberal arts rather than STEM fields. These patterns reveal structural gaps that may constrain long-term cybercrime scholarship and highlight the need for stronger curricular integration and interdisciplinary faculty development.

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Cybercrime education; doctoral training; criminology and criminal justice; curriculum design; interdisciplinary integration

Introduction

Advances in information technology and artificial intelligence have fundamentally transformed the nature of crime, extending criminal activity beyond physical spaces into increasingly complex digital environments (Wu et al., 2025). Data from the Federal Bureau of Investigation's Internet Crime Complaint Center (2025) indicate that 859,532 cybercrime complaints were reported in 2024, with estimated financial losses exceeding 16 billion dollars, a 33% increase from the previous year. These trends underscore the rapid growth and escalating societal impact of cybercrime, positioning it as a distinct and consequential domain within criminology and criminal justice.

In response, a substantial body of scholarship has emerged addressing cybercrime's conceptual foundations, classification systems, prevention strategies, regulatory frameworks, law enforcement responses, and policy implications (Chawki, 2005; Donalds & Osei-Bryson, 2019; Dupont, 2019; Gordon & Ford, 2006; Holt, 2018; Holt & Bossler, 2015; Jahankhani et al., 2014; Loader & Thomas, 2013; Moitra, 2005; Nurahman, 2020;

Phillips et al., 2022; Tropina & Callanan, 2015). While this literature has significantly advanced theoretical and applied understanding of cybercrime, far less attention has been paid to the institutional foundations that sustain this research agenda, particularly the role of doctoral education in preparing scholars to study and teach cybercrime.

Doctoral programs serve as the primary pipeline for knowledge production, faculty development, and long-term disciplinary growth (Altbach, 2007). As cybercrime continues to expand in scope and technical complexity, doctoral training plays a critical role in shaping the field's capacity to generate rigorous scholarship, integrate interdisciplinary methods, and educate future practitioners and researchers. Despite this central role, there is limited empirical evidence on how cybercrime is incorporated into criminology and criminal justice doctoral curricula or whether existing faculty expertise is sufficient to support advanced training in this area.

This study examines cybercrime education in U.S. criminology and criminal justice doctoral programs through two key dimensions: curricular design and faculty capacity. Drawing on a newly constructed database of cybercrime-related courses and faculty across 43 leading doctoral programs, the study assesses how cybercrime is institutionalized within doctoral education and identifies emerging patterns and structural limitations. By analyzing course offerings, program requirements, and faculty training backgrounds, this study provides a systematic overview of the status of cybercrime education in the discipline.

Specifically, the study addresses three research questions: (1) What are the overall trends in cybercrime education within criminology and criminal justice doctoral programs with respect to course design and faculty resources? (2) How is cybercrime education structured across institutions, and what variation exists in course content and organizational placement? (3) To what extent do faculty members specializing in cybercrime possess formal training in this area, and what are their disciplinary backgrounds and career stages?

By addressing these questions, the study contributes empirical evidence to ongoing discussions about doctoral training, interdisciplinary integration, and faculty development, offering insights for scholars, administrators, and policymakers seeking to strengthen cybercrime education within criminology and criminal justice. In addition, integrating cybercrime education into the broader liberal arts curriculum can equip students in this field with not only technical knowledge but also critical thinking skills and cross-disciplinary ethical reasoning. Furthermore, within Criminology and Criminal Justice departments, embedding cybercrime education can help students and scholars adapt to the evolving nature of digital crime, ensuring that the next generation of criminal justice professionals is adequately prepared to address the complexities of twenty-first-century law enforcement and policymaking.

Related work

Cybercrime and cybersecurity

Since the emergence of the term cybercrime and cybersecurity, scholars and institutions have proposed varying definitions to capture its scope and complexity.

Wall (2001) defines cybercrime as the occurrence of a harmful behavior that is somehow related to a computer. Gordon and Ford (2006) define cybercrime as any crime that is facilitated or committed using a computer, network, or hardware device. Loader and Thomas (2013) describe cybercrime as computer-mediated activities which are either illegal or considered illicit by certain parties and which can be conducted through global electronic networks. The U.S. Department of Homeland Security (2020) defines cybersecurity as an activity or process that protects and/or defends information and systems against damage, unauthorized use or modification, or exploitation. Recently, the National Academies of Sciences Engineering, and Medicine (2025) published a consensus panel report on cybercrime classification and measurement, conceptualizing “cybercrime” as crime with a cyber/computer component rather than as cyberactivity with a criminal component. The report further categorizes cybercrime into three types: cyber-dependent crime/pure cybercrime, cyber-enabled crime/hybrid cybercrime, and cyber-assisted crime/computer-assisted traditional crime. These definitions illustrate that the concept of cybercrime and cybersecurity are inherently linked to the use of computers and networks, and taking measures to protect cyber security can effectively reduce the occurrence of cybercrime.

Cybercrime and cybersecurity exhibit clear interdisciplinary characteristics spanning computer science, the social sciences, and other related fields—a consensus widely acknowledged among scholars. For example, Payne and Hadzhidimova (2018) argue that cybersecurity is an interdisciplinary pursuit that encompasses both Science, Technology, Engineering, and Mathematics (STEM) disciplines and the social sciences. Jaishankar (2018) contends that cybercrime constitutes a multidisciplinary field of inquiry involving criminology, sociology, psychology, victimology, information technology, and computer science. Similarly, Wu et al. (2023) argue that cybercrime and cybersecurity involve both human and technological dimensions, requiring investigation through an interdisciplinary lens. Overall, the interdisciplinary nature of these fields necessitates collaboration among experts from diverse domains, including criminologists, computer scientists, cybersecurity specialists, and legal scholars. However, at the same time, this complexity poses significant challenges for curriculum development and educational practice in the realm of cybercrime and cybersecurity.

Cybercrime education and faculty development

In recent years, higher education institutions across various countries and regions have increasingly recognized the importance of cybercrime and cybersecurity, introduced specialized cybercrime- and cybersecurity-related programs, and accumulated valuable teaching experience in instructional practice. This global trend reflects the urgent need to prepare students for the complex and evolving landscape of digital threats, while simultaneously addressing the interdisciplinary challenges inherent in cybercrime education.

In the past decade, numerous universities in the United States have launched certificate programs, academic minors, and full degree offerings in cybercrime and cybersecurity, within criminal justice and criminology programs. For example, Georgia Southern University and the University of South Florida offer specialized certificates

as well as undergraduate and graduate degree programs in cybercrime (Bossler & Berenblum, 2019). Florida State University offers a dedicated major in Cyber Criminology (Jaishankar, 2018). The University of Alabama at Birmingham's Master of Science in Cybersecurity program includes courses such as Cyber Defense and Cyber Crime Investigations (Windsor, 2021). Internationally, several institutions have also made significant contributions to cybercrime and cybersecurity education. In China, East China University of Political Science and Law offers a course titled *Cybercrime and Computer Forensics*, specifically designed for liberal arts students with limited knowledge of computer science and minimal technical background (Tang, 2017). This approach reflects an effort to broaden access to technical topics among students in non-STEM disciplines. In Spain, the Universitat Jaume I (UJI) of Castellón offers a Criminology degree that includes a course titled *Cybersecurity and Cybercrime*, which integrates both theoretical and practical aspects of information and communication technologies (Mollar & Chalmeta, 2016). In Germany, Eberhard Karls University Tübingen's Faculty of Law, the University of Mannheim's Laboratory for Distributed Systems and the University of Applied Sciences Albstadt-Sigmaringen jointly offer a master's program in Digital Forensics, covering topics such as network forensics, digital forensic methodologies, and legal issues in both criminal and civil contexts (Rieger & Kovács, 2010).

The examples above illustrate that cybercrime and cybersecurity programs are primarily concentrated at the undergraduate and master's levels, with relatively few doctoral programs offering specialized degrees or coursework in cybercrime. Doctoral engagement with the field remains limited and uneven across institutions. In most cases, cybercrime courses are housed within liberal arts units such as criminology or criminal justice, whereas cybersecurity curricula are more commonly situated within engineering, computer science, or information technology departments. This institutional separation reflects the dual nature of the field, with cybercrime conceptualized as a social phenomenon and cybersecurity approached as a technical domain. Despite this structural divide, the interdisciplinary nature of cyber-related issues generates substantial overlap in instructional content. For instance, students in cybercrime programs to some extent need to complete foundational computer science coursework such as information systems and data structures, to develop the technical competencies necessary for advanced criminological training.

Beyond disciplinary boundaries, cybercrime higher education in the United States is shaped by broader institutional and workforce frameworks, which in turn further influence course development and faculty specialization. For example, the development of criminal justice curricula is, to some extent, guided and constrained by quality standards established by professional associations such as the Academy of Criminal Justice Sciences (ACJS). In addition, although national-level initiatives such as the National Initiative for Cybersecurity Education (NICE) Cybersecurity Workforce Framework (Newhouse et al., 2017) aim to strengthen the cybersecurity workforce pipeline, their predominantly technical orientation may not fully align with criminological approaches.

From a faculty development perspective, Jaishankar (2018) argues that contemporary cybercrime education is fragmented across three instructional domains: cybersecurity and digital forensics, cyber law, and cyber criminology. Many scholars have raised similar concerns regarding the integration of cybersecurity into criminal justice

education. Belshaw et al. (2020) find that specialized cybersecurity offerings, including courses on the Dark Web and cyber investigations, remain limited within criminal justice curricula. Nodeland et al. (2019) likewise report that most cybersecurity degree programs are situated in computer science, engineering, or business units rather than in criminal justice departments. Payne and Hadzhidimova (2018) argue that criminal justice programs must balance socio-legal foundations with the growing demand for technically oriented cybersecurity competencies. In addition, Krupa et al. (2025) note that the development of cybercrime training for frontline officers is often constrained by state-level training approval processes and resource limitations. This segmentation reflects the broader fragmentation of cybercrime instruction, which often lacks a unified theoretical foundation or cohesive pedagogical framework. Despite these challenges, universities, particularly in the United States, have increasingly begun to recruit tenure-track faculty specializing in cybercrime, most often at the assistant professor level. This emerging hiring trend signals growing institutional recognition of the academic and societal importance of cybercrime and suggests a longer-term commitment to building scholarly capacity and educational infrastructure in this evolving field.

However, the existing literature remains heavily skewed toward workforce development in cybersecurity rather than scholarly exploration of cybercrime education in liberal arts contexts. For example, Burrell (2019) calls for expanded doctoral-level faculty training and proposes the creation of an 18-credit postdoctoral diploma bridge program in cybersecurity to meet workforce demand. While such proposals are valuable, they primarily address the STEM-centric side of the discipline and largely overlook the social science-oriented study of cybercrime.

There remains a notable gap in the literature regarding the academic profiles, doctoral training institutions, disciplinary backgrounds, and hiring networks of faculty who specialize in cybercrime. This lack of systematic evidence limits understanding of the faculty ecosystem shaping this emerging field and constrains efforts to assess institutional capacity, curriculum development, and scholarly productivity in cybercrime education, particularly within liberal arts contexts. As a result, there is a critical need for research that maps patterns of faculty expertise, hiring trajectories, and pedagogical practices to inform the future development of cybercrime and cybersecurity education across institutions of higher education.

Methods

This study focuses on doctoral programs in criminology and criminal justice due to their central role in shaping the future of the discipline. These programs serve not only hubs for advanced research and curriculum development and innovation but also as critical training grounds for the next generation of faculty, scholars, and academic leaders. The program profile includes 43 criminology doctoral programs, from 42 top-ranked graduate schools, ranked by *U.S. News & World Report* in 2021, as well as the University of Alabama. While many excellent undergraduate and master's programs make vital contributions to criminology education and professional practice, doctoral programs provide a distinct vantage point for analyzing long-term trends in faculty expertise, course design, and institutional priorities. As such, they offer a

Table 1. The description of data sources

	Data source	Online source
1	Ranking of graduate schools for a doctorate in criminology (<i>U.S. News</i>)	Website: https://www.usnews.com/best-graduate-schools/top-humanities-schools/criminology-rankings
2	Database of selected courses related to cybercrime	U.S. doctoral programs' website in criminology/criminal justice (e.g. Criminal justice doctoral program at Michigan State University: https://cj.msu.edu/graduates/doctoral.html)
3	Database of faculties with cybercrime backgrounds	Faculties' personal academic website (e.g. Dr. George Burruss: https://www.usf.edu/cbcs/criminology/faculty-staff/g-burruss.aspx)

meaningful and representative sample for understanding how criminology is evolving at the highest levels of academic training in the United States.

The data sources are outlined in Table 1. Research data were collected manually from the official websites of doctoral programs in criminology or criminal justice at these institutions, as well as from the personal academic webpages of faculty members specializing in cybercrime in 2025. We applied the following selection criteria: if one university's criminology program included cybercrime-related courses and faculty members with relevant backgrounds, it was included in the study. Specifically, a cybercrime-related course is defined as a course whose title and description contained at one of the following keywords: "cyber," "digital," "computer," "internet," "technology," "information," or "data."¹ We then manually reviewed specific course titles, descriptions, and instructors, ensuring the relevance to cybercrime. For intelligence courses, only those that explicitly addressed cyber-intelligence were included. Finally, we compiled the collected course information into a database of cybercrime-related courses. In addition, we identified faculty member² with expertise in cybercrime and cybersecurity and collected data such as doctoral graduation year, awarding discipline, alma mater, and research interests, which were organized into a faculty database.

It is important to note that, in the domain of cybersecurity, the National Centers of Academic Excellence in Cybersecurity (NCAE-C) designation system represents the most prestigious form of validation for cyber-related academic programs in the United States. The system is jointly managed by the National Security Agency (NSA) and the Cybersecurity and Infrastructure Security Agency (CISA). It oversees a collaborative cybersecurity education initiative involving 498 community colleges, colleges, and universities and classifies participating institutions into three designation types: Cyber Defense (CAE-CD), Cyber Research (CAE-R), and Cyber Operations (CAE-CO) (National Security Agency, [n.d.](#)). Despite the authority of this designation, the NCAE-C system primarily emphasizes computer science-oriented technical proficiency in cyber defense and operations, which often falls outside the social science-based curricula of

¹We note that the timing of when institutions developed and began offering cybercrime-related courses varies, and this information is not always available online. Depending on departmental needs and circumstances, course names may not necessarily reflect institutions' strength and faculty profiles.

²We note that the identification of cybercrime-related faculty was conducted through a comprehensive assessment of their research interests, publications, and institutional roles, as presented in publicly available departmental profiles. An inclusive approach was adopted to capture the broader landscape of faculty engagement in cybercrime-related research and development.

Criminology and Criminal Justice departments. Accordingly, this study does not adopt the NCAE-C designation framework, to avoid obscuring the disciplinary focus and findings.

On the other hand, given the interdisciplinary nature of cybercrime and cybersecurity, scholars affiliated with departments such as sociology, anthropology, psychology, social work, or related fields frequently conduct research on cybercrime topics. Also, some institutions do not have criminology and criminal justice departments; in such cases, faculty who teach and conduct research on cybercrime are typically housed in sociology or related fields. For example, Dr. Kevin Steinmetz in the Department of Sociology, Anthropology, and Social Work at Kansas State University and Dr. Catherine Marcum in the Department of Government and Justice Studies at Appalachian State University both have active research agendas in cybercrime. However, to assess the internal capacity of criminology and criminal justice doctoral programs specifically, this study excludes interdisciplinary programs and adjacent departments, thereby maintaining clarity and consistency in the analysis.

After data collection and processing were completed, this article draws upon these databases to analyze the current state of doctoral education in cybercrime. To address the research questions posed, the analysis is structured into three main components: (1) Overall Trends: examining course design, faculty recruitment, and research laboratory related to cybercrime in doctoral education; (2) Course Design: analyzing course offerings in cybercrime and cybersecurity at both doctoral and master's levels, including distinctive programs; and (3) Faculty Resources: exploring the academic backgrounds, degree awarding field, and faculty hiring network of faculty members in cybercrime. Two independent coders reviewed the courses and faculty information.

Findings

Overall trends

By reviewing the course and faculty catalogues of each university³, we identified 21 universities offering graduate-level cybercrime-related courses and employing faculty engaged in this field (Figure 1). Overall, universities with cybercrime-related courses and faculty accounted for 48.83% of the total sample ($N=43$), indicating that graduate education in criminology, particularly in emerging areas such as cybercrime and cybersecurity, remains in an early stage of development. More than half of the universities have yet to offer graduate-level courses related to cybercrime or appoint faculty members with relevant expertise. As shown in Figure 1, we categorize the universities into three groups based on the count of cybercrime-related courses and faculty: (a) Strong foundation in cybercrime education, including the University of South Florida, Sam Houston State University, University of

³It should be noted that the data collection and processing was conducted in June 2025. Therefore, we consider the data and information on the official websites of universities and programs prior to this date to be the most recent available. In order to maintain data stability and sustainability, subsequent analyses in this paper do not take into account factors such as changes in course and faculty catalogues over time.

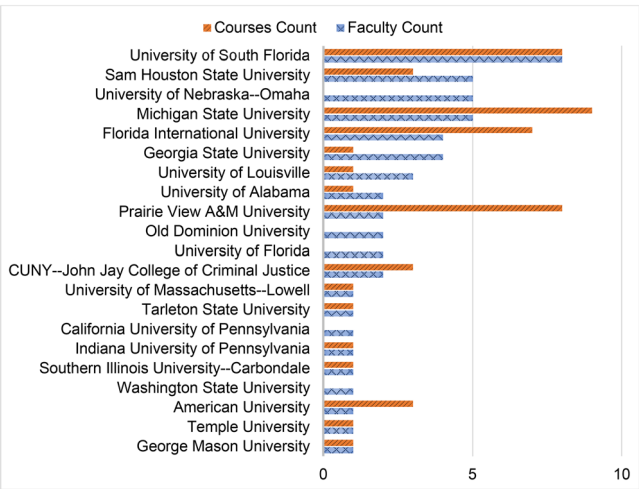


Figure 1. Count of graduate-level cybercrime related courses and faculty at different universities.

Table 2. Information of key cybercrime related labs

University name	Key cybercrime related lab	Director
CUNY-John Jay College of Criminal Justice	Center for Cybercrime Studies	Dr. Marie-Helen Maras
Georgia State University	Evidence-Based Cybersecurity Research Group	—
Michigan State University	The Terrorism, Extremism, and Cybercrime (TEC) Lab	—
Temple University	Cybersecurity in Application, Research, and Education (CARE) Lab	Dr. Aunshul Rege
University of South Florida	The Cybercrime Interdisciplinary Behavioral Research Laboratory (CIBR Lab)	Dr. George Burruss
University of Nebraska-Omaha	National Counterterrorism Innovation, Technology, and Education Center (NCITE)	Dr. Gina Ligon

Nebraska—Omaha, Michigan State University, Florida International University, Georgia State University, and the University of Louisville; (b) Moderate foundation in cybercrime education, including the University of Alabama, Prairie View A&M University, Old Dominion University, University of Florida, and CUNY—John Jay College of Criminal Justice; and (c) Limited foundation in cybercrime education, including University of Massachusetts Lowell, Tarleton State University, California University of Pennsylvania, Indiana University of Pennsylvania, Southern Illinois University—Carbondale, Washington State University, American University, Temple University, and George Mason University.

In addition, the establishment of key laboratories and research centers plays a vital role in advancing both teaching and research in criminology. We identified six well-established laboratories in the field of cybercrime (Table 2), which represent the current leading efforts in this domain. These include the Cybercrime Interdisciplinary Behavioral Research Laboratory (CIBR Lab) at the University of South Florida; the Terrorism, Extremism, and Cybercrime (TEC) Lab at Michigan State University; the National Counterterrorism Innovation, Technology, and Education Center (NCITE) at the University of Nebraska—Omaha; the Evidence-Based Cybersecurity Research Group at Georgia State University; the Center for Cybercrime

Studies at CUNY–John Jay College of Criminal Justice; and the Cybersecurity in Application, Research, and Education (CARE) Lab at Temple University. It is also important to note that these laboratories differ in terms of their founding entities and resource structures. For example, some are established as collaborative research platforms affiliated with Criminal Justice departments, such as the TEC Lab at Michigan State University. Others are faculty-led laboratories centered around individual researchers, such as the CARE Lab at Temple University. Still others are supported by national-level funding initiatives, such as the NCITE Center at the University of Nebraska–Omaha.

Course design

By checking the course requirements of doctoral programs in Criminology or Criminal Justice across universities, we compiled a summary of cybercrime-related elective courses in PhD education (Table 3). In general, current doctoral programs treat cybercrime as a specialized elective course within their curricular frameworks rather than a core course requirement. This indicates that cybercrime—an emerging and specialized area—remains a peripheral branch of criminology education and has yet to become a central component of doctoral curricula. For instance, in the Criminal Justice doctoral program at the University of Louisville⁴, students are required to complete eight core courses (e.g. *CJ 745: Advanced Criminal Behavior*, *CJ 748: Seminar in Criminal Justice Program Planning & Evaluation*) and 18 credit hours of electives. Among these, only one 3-credit course specifically addresses cybercrime (*CJ 627: Computer and Cyber Crime*).

In terms of course count, Michigan State University, CUNY–John Jay College of Criminal Justice, and American University demonstrate significant emphasis on cybercrime instruction, each offering more than three cybercrime-related electives. In contrast, several other institutions—such as Florida International University, Georgia State University, University of Louisville, University of Massachusetts Lowell, University of Alabama, George Mason University, Temple University, Southern Illinois University–Carbondale, Indiana University of Pennsylvania, and Tarleton State University—offer only a single cybercrime course, often framed as a specialized topic.

In terms of course content, most universities offering only a single cybercrime elective provide a general overview of the field, with limited thematic diversity. For example, the University of Louisville's *CJ 627: Computer and Cyber Crime* covers legal, policy, and technical issues associated with computer, digital, and electronic crime, including digital forensics, information security, and privacy. In contrast, Michigan State University offers a broader array of cybercrime electives. Beyond introductory content, it provides targeted courses on data infrastructure, open-source analysis, and cyber terrorism, such as *CJ 872: Open Source Information Analysis*, *CJ 874: Cybercrime, Deviance and Virtual Society*, *CJ 876: Data Systems/Infrastructure*, and *CJ 877: Cyber Terror and Cyber Warfare*.

⁴<https://catalog.louisville.edu/graduate/programs-study/doctor-philosophy-criminal-justice/#degreerequirementtext>.

Table 3. Information of cybercrime related elective courses in PhD education

University name	Cybercrime related elective courses	Count
Michigan State University	CJ 872 Open Source Information Analysis CJ 874 Cybercrime, Deviance and Virtual Society CJ 876 Data Systems/Infrastructure CJ 877 Cyber Terror and Cyber Warfare	4
CUNY--John Jay College of Criminal Justice	ICJ 760 Information Technology and Cybercrime CRJ 727 Cyber criminology CRJ 753 Investigating Cybercrime	3
American University	JLC 496 Selected Topics: Cybercrime and Public Policy JLC 445 Topics in Criminology Cyber criminology JLC 683 Cyber Threats and Security	3
Florida International University	CJE 4694 Cyber Crime	1
Georgia State University	CRJU 8900 Cyber Crime Investigations	1
University of Louisville	CJ 627 Computer and Cyber Crime	1
University of Massachusetts Lowell	CRIM 6580 Issues in Computer Crime and Cyber Security	1
University of Alabama	CJ 590 Special Topics: Cybercrime	1
George Mason University	CRIM 795 Cybercrime	1
Temple University	CJ 8310 Privacy, surveillance, and Cybercrime	1
Southern Illinois University--Carbondale	CCJ 325 Special topics: in Cybercrime	1
Indiana University of Pennsylvania	CRIM 645 The Dynamics of Cybercrimes	1
Tarleton State University	CRIJ 5353 Global Cyber-Security	1

In addition to doctoral programs, we also identified unique master's and minor programs related to cybercrime and cybersecurity in our sample (Table 4). For example, Michigan State University offers an online Master's degree program in Cybercrime and Digital Investigation⁵; the University of South Florida provides an online Master of Science in Cybercrime program⁶; and Florida International University offers a Master of Arts in Global Affairs, with a concentration in Cybersecurity and Technology Policy⁷, housed within the Steven J. Green School of International and Public Affairs. Additionally, Prairie View A&M University features an interdisciplinary minor in Cybersecurity, allowing students to select cybersecurity-related courses based on their individual interests and career goals. Also, the University of Massachusetts Lowell has a Master of Science (M.S.) program in Security Studies with concentration in cybersecurity.⁸

Furthermore, due to the interdisciplinary nature of cybercrime, research and instruction in this area are not confined to traditional departments of criminology and criminal justice. A notable example is Sam Houston State University, home to one of the largest and most prestigious criminal justice programs in the United States. Its College of Criminal Justice comprises four departments: the Department of Criminal Justice and Criminology, the Department of Forensic Science, the Department of Security Studies, and the Department of Victim Studies. Our review found that faculty members specializing in cybercrime and cybersecurity are also located within the Department of Security Studies and the Department of Victim Studies. These

⁵<https://online.cj.msu.edu/ms-masters-cybercrime-digital-investigation/curriculum>.

⁶<https://www.usf.edu/cbcs/criminology/graduate/cybercrime/>.

⁷<https://sipa.fiu.edu/global-affairs/program/cyber-security-policy/index.html>.

⁸<https://www.uml.edu/fahss/security-studies/programs/program-overview.aspx>.

Table 4. Information of other cybercrime featured programs and courses

University name	Other cybercrime featured programs and courses	Count
Michigan State University	(Master's Degree in Cybercrime and Digital Investigation) CJ 881 Legislative and Policy Responses to Cybercrime CJ 882 Analysis of Contemporary Cyberthreats CJ 875 Digital Forensic Investigations CJ 878 Economic Cybercrimes and Fraud CJ 879 Interpersonal Cybercrime	5
University of South Florida	(Online Master of Science in Cybercrime) CCJ 6057 Cybercrime Capstone CJE 6688 Cybercrime and Criminal Justice CCJ 6602 Profiling Cybercrime CJE 6690 Cybercrime Law and Social Policy CCJ 6637 Technology Adoption & Crime CJE 6698 Cyber Victimology CJE 6627 Digital Evidence Recognition and Collection CJE 6626 Digital Forensic Criminal Investigations	8
Florida International University	(Master of Arts in Global Affairs-Cybersecurity and Technology Policy program) CIS 5027 Computer Systems Fundamentals ISS 5654 Foundations of Cybersecurity and Technology Policy ISS 5656 Cyberspace and Globalization ISS 6655 Issues in Cybersecurity and Technology Policy ISS 6658 Cyber Warfare and Strategy LAW 7707 Cybersecurity and Privacy Law	7
Prairie View A&M University	(Cybersecurity Interdisciplinary Minor) CRIJ 2381 Fundamentals of Cybersecurity GNEG 4352 Advanced Fundamentals of Cybersecurity MISY 3332 Networking CRIJ 3348 Cyber Terrorism and Cyber Defense CRIJ 4347 Digital Forensics Investigations GNEG 4350 Cybersecurity and Public Policy MISY 4332/CRIJ 4333 Enterprise Cybersecurity PHIL 2309 Ethics of Cybersecurity	8
Sam Houston State University	(Department of Security Studies) SCST 5348 Critical Infrastructure Protection SCST 6364 Cybersecurity (Department of Victim Studies) VCST 5372 Cybervictimization	3

departments additionally offer courses related to cybersecurity (*SCST 6364 Cybersecurity*) and cyber victimization (*VCST 5372 Cybervictimization*) as elective options for students.

Faculty resources

Among the 43 universities sample in this study, a total of 54 faculty members with academic backgrounds in cybercrime were identified. This following section analyzes the distribution of their doctoral graduation time, academic positions, degree-conferring fields, and hiring flow patterns to better understand faculty development trends in the field of cybercrime.

It is important to note that, given the interdisciplinary and complex nature of cyber-crime studies, as well as the field's future development and the flexible career paths of faculty members, we have adopted a relatively broad definition in identifying "cyber" scholars. Specifically, in addition to faculty members who explicitly indicate cybercrime as a research area on their personal websites or curricula vitae, we also include those whose expertise lies in closely related domains such as cybersecurity, cyberbullying and

cybervictimization. Moreover, scholars whose primary research may not center on cybercrime, but who have participated in cyber-related research projects or coauthored publications on cyber topics are also categorized as “cyber” scholars in this paper. In addition, we have also considered the faculty appointments within each department. As the development of cybercrime disciplines is dynamic, rigid definitional boundaries may hinder progress. Thus, the inclusive definition adopted in this study offers a more comprehensive representation of faculty expertise in the cyber domain.

Faculty doctoral graduation time

The analysis of faculty doctoral graduation time (Figure 2) illustrates a clear upward trend in the number of recent doctoral graduates among current university faculty. Only three faculty members earned their doctorates before 2000, with increasing numbers in each subsequent period: 2000–2004 ($N=4$), 2005–2009 ($N=5$), 2010–2014 ($N=7$), 2015–2019 ($N=7$), and 2020–2024 ($N=10$). This suggests a growing interest in cybercrime-related research among newer generations of faculty, with the majority having earned their degrees within the last 10–15 years. The notable rise from 7 to 10 faculty members between the 2015–2019 and 2020–2024 cohorts indicates active hiring of recent PhDs, likely driven by increasing demand for expertise in cybercrime and the expansion of related academic programs.

Faculty academic positions

We examined the academic positions of each faculty member and constructed a bar chart to visualize the number of faculty at different career stages (Figure 3). Among the 53 faculty members identified, 9 are full professors, representing 16.98% of the total. There are 20 associate professors (37.74%) and 19 assistant professors (35.85%), while 5 faculties (9.43%) hold other positions such as lecturer, research associate, or instructor.

Classified by career stage, 16.98% of faculty members are in the late-career stage (professors), 37.74% are in the mid-career stage (associate professors), and 45.28% are in the early-career stage (assistant professors and equivalent positions). This distribution indicates that cybercrime is an emerging and developing discipline, with ongoing needs for faculty recruitment to strengthen teaching and research capacity. Furthermore, early-career scholars, particularly assistant professors, require increased

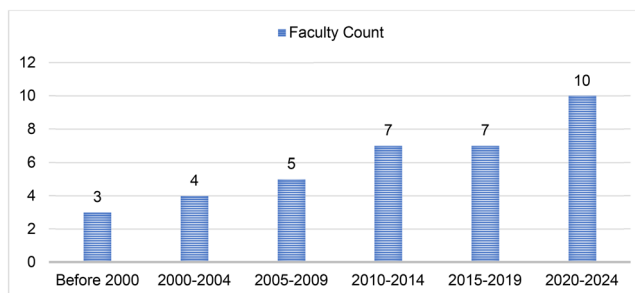


Figure 2. Faculty count of doctoral graduation time.

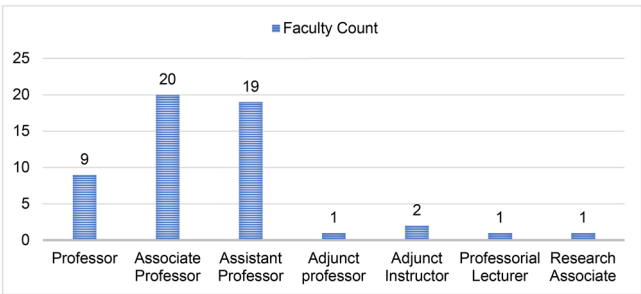


Figure 3. Faculty count of academic positions.

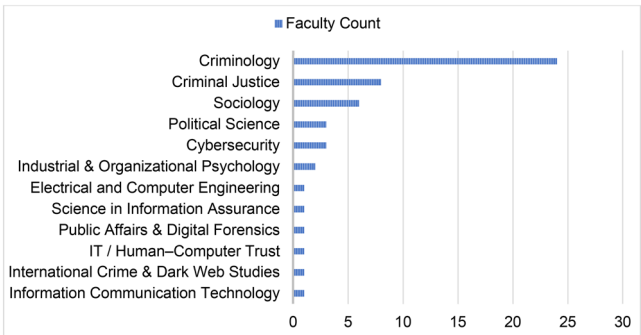


Figure 4. Faculty count of doctoral degree fields.

research support and resource allocation to foster their professional development and advance the field of cybercrime.

Faculty doctoral degree fields

The analysis of faculty doctoral degree fields (Figure 4) reveals a strong concentration in several core academic disciplines relevant to cybercrime and criminal justice. Criminology is by far the most common field, accounting for 24 faculty members. It is followed by Criminal Justice, with 8 faculty members sharing similar educational backgrounds, and Sociology, with 6 faculty members. Collectively, these three disciplines account for approximately 71.70% of all faculty doctoral degrees, highlighting the dominance of social science disciplines in cybercrime education. This pattern also reflects the traditional academic foundations of cybercrime research and teaching.

Emerging and interdisciplinary fields are also represented, though to a lesser extent. Political Science and Cybersecurity each contribute three faculty members, suggesting that the integration of policy and technical perspectives is having a growing impact on cybercrime education. In addition, a small number of faculty members come from related areas such as Industrial and Organizational Psychology, Electrical and Computer Engineering, and Information and Communication Technology, each represented by one to two faculty members. These fields bring important yet often underrepresented perspectives to the broader field of digital crime studies.

Overall, the data indicate that while cybercrime programs remain rooted in criminology and other social science disciplines, the increasing inclusion of faculty from technical and interdisciplinary backgrounds should not be overlooked, as they constitute a meaningful proportion of the academic workforce. This trend may reflect the growing complexity of cybercrime challenges and the need for multifaceted academic approaches.

Faculty hiring flow

To explore the academic mobility of faculty with a background in cybercrime following the completion of their doctoral degrees, we collected information on each faculty's doctoral-granting origin and current academic employer. A Sankey diagram was constructed to visualize faculty hiring flow (Figure 5). As illustrated in the diagram, the leading institutions producing cybercrime PhD faculty include the University of South Florida ($N=4$), Michigan State University ($N=4$), University of Missouri–St. Louis ($N=3$), Florida State University ($N=3$), Indiana University of Pennsylvania ($N=3$), University of Oklahoma ($N=2$), University of Cincinnati ($N=2$), and Florida International University ($N=2$). Notable institutions also represented include New York University, the University of Florida, the Ohio State University, the University of Georgia, the University of Maryland, Rutgers University, and Northwestern University.

Regarding faculty recruitment, the top five universities employing faculty specializing in cybercrime are the University of South Florida ($N=8$), Michigan State University ($N=5$), University of Nebraska Omaha ($N=5$), Sam Houston State University ($N=5$), and Georgia State University ($N=4$). This pattern suggests that universities actively developing programs in cybercrime and cybersecurity possess both a strong motivation and competitive advantage in attracting faculty with relevant expertise.

It is also common for faculty to remain employed at their doctoral-granting institutions. For instance, three doctoral graduates from the University of South Florida continue to serve at the institution, contributing to cybercrime teaching and research. Similarly, one faculty member each from Washington State University and Florida International University remain at their respective institutions.

Additionally, the data reveal that several faculty members obtained their doctorates from international institutions. Notable examples include the University of Cambridge in the United Kingdom, Tel Aviv University and the Hebrew University of Jerusalem in Israel, Simon Fraser University in Canada, and the National University of Singapore. This pattern suggests that the faculty composition in the United States exhibits a degree of international diversity. Nevertheless, further efforts to attract global talent remain essential for advancing the development of cybercrime as an academic discipline.

Discussion

Drawing on data from 43 leading doctoral programs in criminal justice in the United States, this study identified institutions, courses, and faculty associated with cybercrime, establishing a comprehensive database of related courses and faculty members. The research examined current trends in cybercrime education in the United States

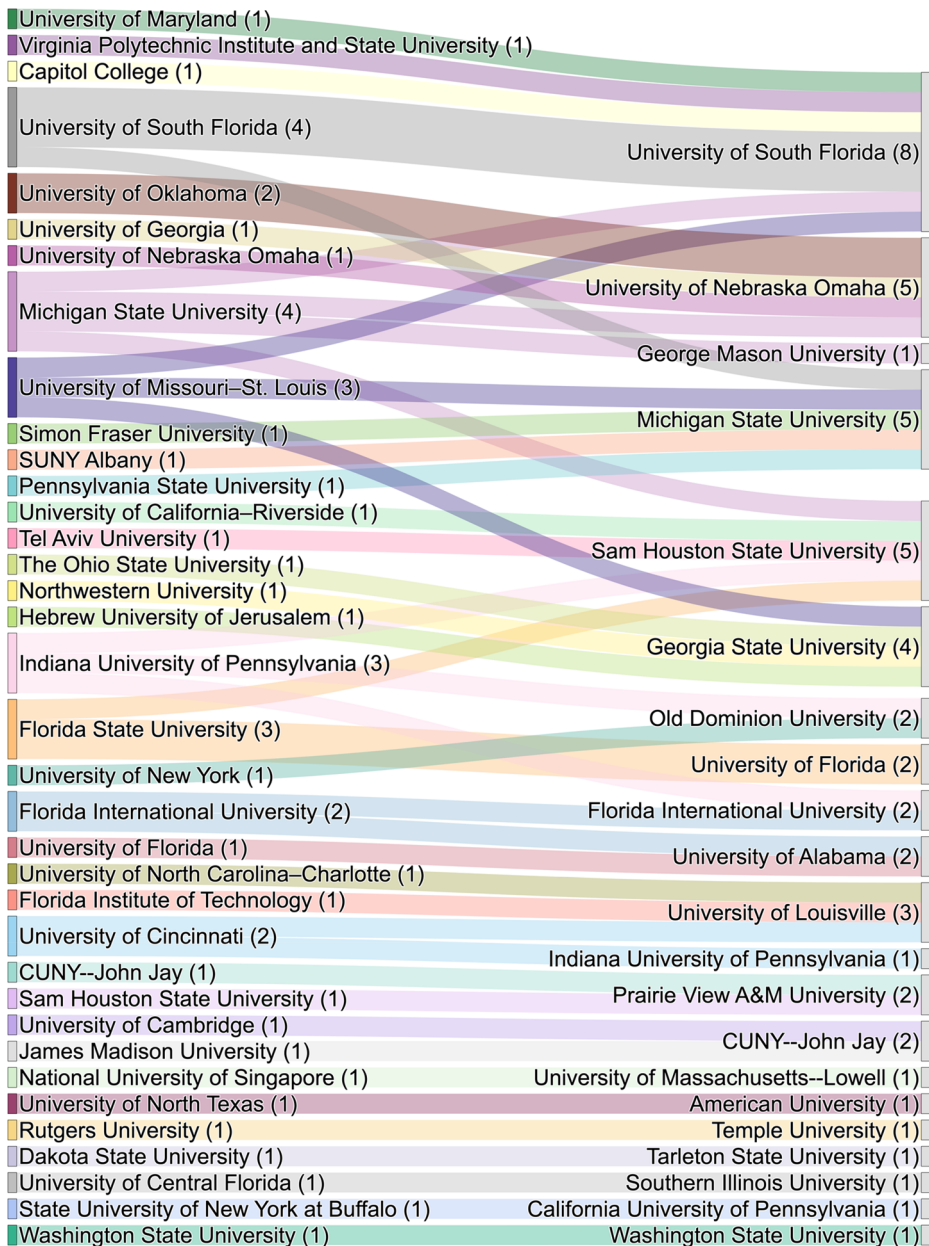


Figure 5. Faculty hiring flow in cybercrime.

through the dual lenses of course design and faculty resources. The key findings reveal three critical patterns.

First, nearly half of the surveyed universities offer cybercrime-focused courses with dedicated faculty. Only six laboratories and research centers dedicated to cybercrime have been established. The institutions demonstrating strongest foundations in this field include the University of South Florida, Sam Houston State University, Michigan State University, the University of Nebraska–Omaha, and Florida International University.

Second, criminology doctoral programs treat cybercrime as a peripheral elective rather than a core curricular requirement. While a few institutions, such as Michigan State University, offer a broader selection of cybercrime courses, most universities provide only general, introductory-level instruction, with course content remaining relatively limited in scope. Moreover, select institutions, including the University of South Florida and Michigan State University, have developed online Master of Science programs in Cybercrime.

Third, approximately 45% of faculty specializing in cybercrime are early-career scholars, with most earning their doctoral degrees within the past decade. A majority (71.70%) of these faculty hold doctorates in liberal arts disciplines such as criminology and criminal justice, while a minority possess STEM backgrounds, including computer science. The primary doctoral origins of these scholars include the University of South Florida, Michigan State University, University of Missouri–St. Louis, Florida State University, and Indiana University of Pennsylvania. International institutions such as the University of Cambridge have contributed to faculty but remain underrepresented.

Furthermore, when considered holistically, a notable finding is the apparent decoupling between institutional resources, such as research laboratories, faculty capacity, faculty teaching/work loads, or master's-level concentrations, and doctoral-level course offerings and training. In other words, the presence of a dedicated cybercrime research laboratory does not necessarily correspond to a greater number of doctoral-level cybercrime courses within a department, and the reverse pattern is also observed. For example, Georgia State University boasts abundant faculty and laboratory resources, but offers only one cybercrime-related doctoral course, *CRJU 8900 Cyber Crime Investigations*. This pattern indicates that institutional resources and doctoral-level training often operate independently in practice. It further suggests that efforts to strengthen doctoral-level cybercrime education require coordinated development across these components rather than treating them as isolated elements.

Cyber criminology is a subfield of criminology that represents a critical frontier for the discipline's future development. Doctoral and graduate education in cybercrime is particularly important for advancing the field and cultivating specialized talent. This study aimed to examine current trends in cyber criminology education by focusing on curriculum design and faculty resources, providing scholars and the education community with an updated overview of the discipline's educational landscape. The limited integration of cybercrime into core curricula, combined with the predominance of liberal arts-trained faculty, reveals a fundamental tension between the technical demands of cybercrime and traditional criminological approaches. This disciplinary disconnect may impede the development of scholars capable of bridging technological and social dimensions of cybercrime.

Several limitations warrant consideration. First, our research sample included only 43 representative universities offering doctoral programs in criminology, excluding master's-only and interdisciplinary programs that may offer innovative approaches to cybercrime education. We primarily included criminology and criminal justice departments in institutions offering doctoral degrees; thus, we unfortunately did not include faculty hired in other related disciplines. While these institutions provide valuable insight into cybercrime education at the doctoral level, this focus may overlook the broader landscape of cybercrime-related instruction occurring at other types of institutions—particularly

those offering master's or professional programs. Second, institutional conditions that are not publicly disclosed—such as faculty teaching loads and internal resource allocations—fell outside the scope of our analysis. Third, our data collection relied on publicly available information from institutional websites and online course catalogues. This approach, while practical and standardized, treats this information at face value and may not capture the nuanced realities of course implementation, instructional depth, or faculty engagement. It also does not reflect how faculty themselves perceive their expertise, roles, and involvement in cybercrime training and curriculum development. Such limitations suggest the potential for discrepancies between official representations and actual pedagogical practice. This can be addressed in future research.

To address these gaps, future research should seek to expand the sample to provide a more comprehensive analysis of disciplinary development. Specifically, future studies should broaden the institutional scope to include a wider range of academic programs, including master's level offerings and interdisciplinary departments. In addition, incorporating both qualitative and quantitative data such as faculty interviews, student surveys, curricular documents, and classroom observations would enable a more nuanced and empirically grounded understanding of how cybercrime education is structured, delivered, and experienced across higher education institutions. Importantly, future research should also adopt longitudinal designs that track the evolution of cybercrime curricula over time, allowing scholars to examine how programs adapt to technological change, emerging threats, policy shifts, and labor market demands. Such longitudinal perspectives would provide critical insight into curricular stability, innovation cycles, and the long-term institutionalization of cybercrime education within higher education.

Building on the interdisciplinary nature of cybercrime and cybersecurity, this study offers several recommendations for disciplinary advancement, focusing on curriculum and faculty development. First, universities with established doctoral programs in criminology or criminal justice should consider expanding their offerings of cybercrime-related electives. Where faculty resources permit, departments should consider integrating courses on computer systems, data structures, and related subjects to enhance students' interdisciplinary competencies. In the absence of faculty resources, departments can collaborate with STEM fields to develop cross-disciplinary courses. Second, criminology or criminal justice departments should actively recruit doctoral graduates specializing in cybercrime, cybersecurity, and related STEM disciplines, including computer science, to strengthen faculty expertise. Proactively recruiting international cybercrime scholars from outside the United States can further enrich the global diversity of faculty and introduce an international perspective into academic programs. Additionally, given that most current cyber criminology faculty are in the early- to mid-career stages, national, state, and institutional bodies should establish supportive policies and targeted funding mechanisms to foster the growth and development of emerging scholars. Such support is vital for ensuring the sustainable and robust advancement of the cyber criminology discipline.

Conclusion

As the digital landscape continues to blur the boundaries between crime and justice, the integration of cybercrime education into criminology and criminal justice programs

represents a pivotal evolution in the field. This shift is not merely additive but transformative, reshaping academic training, research agendas, and the future direction of the discipline. As cyber-enabled and cyber-dependent crimes grow in frequency and complexity, the demand for specialized knowledge, innovative pedagogical approaches, and interdisciplinary collaboration has become increasingly urgent.

Our findings indicate that current efforts remain insufficient. Most programs treat cybercrime as a peripheral elective, 71.7% of faculty lack technical backgrounds, and only six institutions have established dedicated cybercrime research laboratories. Integrating cybercrime content into curricula encourages programs to move beyond traditional criminal justice frameworks by incorporating areas such as digital forensics, cybersecurity, and online deviance. These additions promote interdisciplinary collaboration across fields including computer science, information systems, law, and behavioral sciences, all of which are essential for understanding and addressing the multifaceted nature of cybercrime. As a result, criminology programs may increasingly adopt hybrid educational models that bridge social science and technical domains.

This curricular transformation is also redefining the skill sets expected of graduates. Tomorrow's criminologists will need not only a foundation in theory and research methods but also practical competencies in areas such as cyber risk, data analytics, and digital ethics. These evolving demands are likely to influence faculty recruitment and development, encouraging institutions to seek interdisciplinary expertise and explore new educational pathways, such as dual-degree and certificate programs.

On the research front, the expansion of cybercrime education creates new opportunities for criminological inquiry, including the study of algorithmic policing, ransomware economies, digital surveillance, and dark web activities. Addressing these complex phenomena requires methodological innovation that integrates big data analytics, computational modeling, and digital ethnography. Such approaches increasingly blur traditional boundaries between quantitative and qualitative research, enabling more comprehensive and nuanced analyses of cybercrime and its social impacts.

Finally, this educational shift has broad implications for public policy and international collaboration. As cybercrime transcends national borders, academic programs that incorporate global perspectives and transnational case studies will be better equipped to prepare students for roles in international law enforcement, cybersecurity policy, and global justice systems. By aligning educational outcomes with workforce and policy demands, criminology programs can strengthen their societal relevance and long-term impact.

Criminology stands at a critical juncture. The discipline must either adapt to the technical and interdisciplinary demands of the digital era or risk marginalization as cybercrime becomes an increasingly central concern in crime and justice systems. By embracing cybercrime education, the field takes an essential step toward future readiness, equipping scholars, practitioners, and institutions with the knowledge and skills necessary to navigate the complex realities of an increasingly interconnected digital world.

Author contributions

CRedit: **Ling Wu**: Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Yuchen Wang**: Data curation, Formal analysis,

Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing; **Claire Seungeun Lee**: Writing – review & editing; **Wei Li**: Writing – review & editing; **James Zhang**: Writing – original draft; **Xinyue Ye**: Conceptualization, Methodology, Resources, Supervision, Validation, Writing – review & editing. Yucheng Wang and Xinyue Ye served as joint corresponding authors.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors employed the OpenAI ChatGPT-5 model as a language-editing aide to enhance readability, grammar, and stylistic clarity. ChatGPT-5 was not used to create, expand, or alter the scientific ideas, data analyses, or conclusions presented herein, and no AI-generated text has been included verbatim in the manuscript.

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